

BERNARDINO BRUNO VITTORINO

SENIOR C++ SYSTEMS ENGINEER | Linux | Distributed Systems | Performance Engineering

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PROFESSIONAL SUMMARY

Senior C++ systems engineer with 18+ years of experience developing performance-sensitive software across distributed backend platforms, telecom systems, network protocols and embedded environments. Strong expertise in modern C++, Linux, multithreading, performance analysis and production debugging. Currently designing and maintaining game-family-specific C++17 backend engines and reusable libraries that implement mathematical models, state machines and transaction workflows while integrating with distributed state, remote services and durable persistence. Experienced in owning complex technical problems from design and implementation through load validation, deployment and production monitoring.

TECHNICAL SKILLS

Languages

C++11/14/17; working knowledge of C++20; C; C#; SQL

Systems

Linux; multithreading; concurrency; distributed systems; client/server; network programming

Data & Integration

Redis; Oracle; SOCI; REST/HTTP; Protobuf; XML; transaction processing

Build, Delivery & Observability

CMake; Git; Jenkins; GitLab CI; TeamCity; Docker; Artifactory; AWX/Ansible; GDB; OpenTelemetry; Dynatrace; Wireshark

PROFESSIONAL EXPERIENCE

Senior Software Engineer (C++), Contract | Entain Group

Aug 2022 - Present

Milan / Remote, Italy

Design and implement game-family-specific C++17 backend engines and libraries containing the mathematical models, algorithms, state machines, payout evaluation, bonus mechanics and transaction workflows used by online slot, roulette and card games.

Own delivery from technical design and implementation through integration, rollout, production monitoring and iteration with QA and operations teams.

Integrate long-running Linux C++ processes with session and RNG services, Redis-based hot state and distributed locking, wallet/account systems, and Oracle persistence accessed through the SOCI C++ abstraction layer.

Analyse critical execution paths and distributed tail latency using controlled load campaigns, Dynatrace, OpenTelemetry, GDB, structured logs and Linux diagnostic tools.

Reduced end-to-end p99 latency by approximately 80% across successive stress baselines and eliminated recurrent reverse-proxy failures by isolating bottlenecks across network, cache, database and downstream services.

Harden retry, compensation and recovery logic to preserve debit, refund, round-state and transaction consistency during timeouts, retries and partial failures.

Contributed to increasing annual backend deliveries from approximately 5 to 25 by creating reusable components and unblocking cross-team integration while maintaining stability and performance targets.

Environment: C++17, Linux, CMake, multithreading, distributed systems, Redis, Oracle, SOCI, REST/HTTP, OpenTelemetry, Dynatrace, GDB, Jenkins, GitLab CI, Artifactory, AWX/Ansible, Git

Senior Software Engineer (C++), Telecom/5G, Contract | Empirix / Infovista

Milan / Remote, Italy

Developed performance-sensitive C++17 components for a distributed and containerised telecom monitoring and testing platform supporting 4G, 5G and VoLTE traffic.

Optimised critical paths by removing avoidable allocations and copies, improving data structures and resolving multithreading, contention and concurrency bottlenecks.

Built repeatable performance baselines and integrated regression thresholds into TeamCity pipelines so builds failed automatically on measurable performance degradation.

Diagnosed failures across distributed Linux and container environments using profiling, targeted reproduction harnesses, logs and protocol-level analysis.

Environment: C++17, Linux/CentOS, multithreading, Docker, Protobuf, GoogleTest, TeamCity, GitLab, CI/CD, Git

Senior Software Engineer (C#/.NET) - Test Automation | Epson Europe

Jun 2019 - Feb 2021

Milan, Italy

Designed and implemented a dynamic C#/.NET automation framework for fiscal printers and payment terminals, covering integration, system, regression, stress and performance testing.

Identified and supported the resolution of 200+ legacy defects, including concurrency, firmware, cryptographic-signing and client/server integration issues.

Built reproducible end-to-end test environments for SSL-based client, web-service and database workflows using NUnit, JMeter and Postman.

Environment: C#/.NET, NUnit, Log4Net, REST APIs, XML, multithreading, JMeter, Postman

EARLIER C++ AND EMBEDDED SYSTEMS EXPERIENCE

C++ Senior Software Engineer | Freelance

Feb 2017 - May 2019

Bologna, Italy

Designed an asynchronous C++11 client/server and REST architecture using cpprestsdk, task-based concurrency and thread pools; also delivered ROS autonomous-navigation and OpenCV computer-vision projects.

Environment: C++11, cpprestsdk, OpenCV, Visual Studio, ROS, Linux, GoogleTest, JMeter

C++ / C Senior Software Engineer | Alten Italia S.p.A.

May 2016 - Jul 2016

Bologna, Italy

Developed a Wireshark dissector for a new SIL4 railway protocol and collaborated with the upstream Wireshark community after identifying defects in the official release.

Environment: C/C++, protocol analysis, reverse engineering, static analysis, Wireshark

C++ Senior Software Engineer | Dream Technologies Ltd

Jan 2011 - Dec 2015

Malta

Developed multithreaded C++ client/server data-processing software and SQL-based analytical components for remote gaming-data collection and player statistics.

Environment: C++11, multithreading, client/server systems, reverse engineering, SQL

Embedded Software Engineer, Contract | Mxengineering

Dec 2008 - Dec 2010

Bologna, Italy

Developed MISRA-C software and a safety-critical SIL4 railway communication protocol on uC/OS for a block axle-counter system.

Environment: C, MISRA-C, uC/OS, safety-critical systems, SIL4, Subversion

C++ Product Engineer | Datalogic S.p.A. - Scanning Group

Oct 2007 - Dec 2008

Bologna, Italy

Developed C/C++ drivers, hardware-test software and high-speed PCI data-acquisition functions for industrial scanning systems, increasing communication throughput from 30 to 75 Mbps.

Environment: C/C++, device drivers, PCI, hardware testing, Borland C++, LabWindows/CVI

Firmware Engineer, Contract | **Magneti Marelli Powertrain S.p.A.**

Jan 2007 - Sep 2007

Bologna, Italy

Developed and tested C firmware for an automotive ECU using white-box, fault-injection and HIL techniques; diagnosed and supported fixes for 200+ firmware defects using CAN tools and laboratory instrumentation.

Environment: C, embedded firmware, ECU, CANoe, CANalyser, CANape, HIL, oscilloscope

EDUCATION AND CERTIFICATION

M.Sc. in Electrical and Computer Engineering | **University of Bologna**

2006

Bologna, Italy

Experimental thesis on Multiprocessor System-on-Chip (MPSoC) architecture, supervised by Prof. Luca Benini.

ISTQB Certified Tester - Foundation Level | 2016